

- HW 1 published

One-time pad

- Let $\mathcal{M} = \{0,1\}^n$
- $\text{Gen}()$: choose a uniform key $k \in \{0,1\}^n$
- $\text{Enc}_k(m)$: $c = k \oplus m$
- $\text{Dec}_k(c)$: $m = k \oplus c$
- Correctness:
$$\begin{aligned}\text{Dec}_k(\text{Enc}_k(m)) &= k \oplus (k \oplus m) \\ &= (k \oplus k) \oplus m = m\end{aligned}$$
- Theorem 1: One-time pad encryption is perfectly secure.

Perfect secrecy

- Algorithms:
 - $k \in \mathcal{K}$: secret key, $m \in \mathcal{M}$: message, $c \in \mathcal{C}$: ciphertext
 - $k = \text{Gen}()$
 - $c = \text{Enc}(k, m)$
 - $m = \text{Dec}(k, c)$
- **Correctness:** for every message $m \in \mathcal{M}$, any secret key $k \in \mathcal{K}$
 $m = \text{Dec}(k, \text{Enc}(k, m))$
- **Perfect secrecy:** for any distribution over the message space $\mathcal{M}$, for every message $m \in \mathcal{M}$ and every ciphertext $c \in \mathcal{C}$,

$$\Pr[M = m \mid C = c] = \Pr[M = m]$$

Lemma:

For every two messages $m_0, m_1 \in \mathcal{M}$ and any ciphertext $c \in \mathcal{C}$,

$$\Pr[\text{Enc}_k(m_0) = c] = \Pr[\text{Enc}_k(m_1) = c]$$

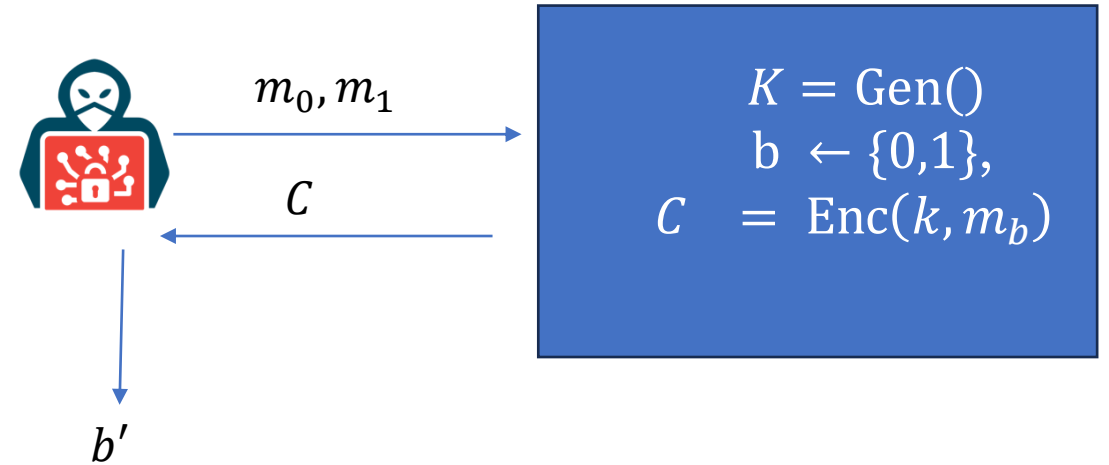
where the probability is over the random choice of k .

An encryption scheme is perfectly secure if and only if the equation above holds

Another def: **perfect indistinguishability**

Experiment:

1. Adversary A chooses any two messages $m_0, m_1 \in \mathcal{M}$
2. Generate a key k using $\text{Gen}()$, pick a random bit $b \in \{0,1\}$, send attacker $c = \text{Enc}(k, m_b)$
3. Attacker guesses b'
4. Experiment outputs succeed if $b = b'$



Perfectly indistinguishable: for every adversary A

$$\Pr[\text{Experiment outputs 1}] = 1/2$$

Weakness of one-time pad: can't reuse the key

- $m_1 \oplus m_2$ is information about m_1, m_2
- Is this significant?
 - $m_1 \oplus m_2$ reveals where m_1, m_2 differ
 - Frequency analysis
 - Exploiting characteristics of ASCII...

Weakness of one-time pad

- Key length has to be the same as message length

A better definition?

- $\mathcal{K} : \{0,1\}^n$
- $\mathcal{M} : \{0,1\}^{n+s}$ or $\mathcal{M} : \{0,1\}^*$?

Not possible!!!

- Shannon's theorem: Size of key space has to be at least the same as the message space $|\mathcal{K}| \geq |\mathcal{M}|$ to achieve perfect secrecy

Computational security

Two relaxations:

1. Security only against **efficient** adversaries.
2. Security can fail with a **very small probability**

Relaxation 1: efficient adversary

Polynomial time: $O(n^c)$ for any constant c

$O(n^3)$ vs. $O(2^n)$

2^{60} seconds since the start of the universe

Security against all probabilistic polynomial time (PPT) adversaries

Relaxation 2: small failure probability

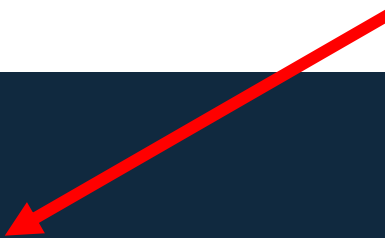
Negligible function f : for every polynomial p , there is an integer N such that for all $n > N$,

$$f(n) < \frac{1}{p(n)}$$

E.g.,: $2^{-n} = \frac{1}{2^n}$, $2^{-\sqrt{n}}$, $n^{-\log n}$

How small is 2^{-60} ?

2^{-14}



[NEWSNATION]

According to the NWS, a person has a 1-in-15,300 chance of getting struck by lightning in their lifetime.

It's said you are four times more likely to get a perfect score on your SAT than getting struck by lightning.

Negligible function $\text{negl}(n)$

$\text{negl}_1(n) + \text{negl}_2(n)$ is negligible

$\text{negl}_1(n) \cdot p(n)$ is negligible for any polynomial p

Definition of computational security

- Algorithm $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$:
 - $k \in \mathcal{K}$ secret key, $m \in \mathcal{M}$: message, $c \in \mathcal{C}$: ciphertext
 - $k = \text{Gen}(\mathbf{1}^n)$
 - $c = \text{Enc}_k(m)$
 - $m = \text{Dec}_k(c)$
- Correctness: $m = \text{Dec}_k(\text{Enc}_k(m))$

n : security parameter
usually use λ

Computation indistinguishability against eavesdropper

Experiment $\text{PrivK}_{A,\Pi}^{\text{eav}}(n)$:

1. Adversary A is given input 1^n , and outputs two messages $m_0, m_1 \in \mathcal{M}$
s.t. $|m_0| = |m_1|$
2. A key k is generated using $\text{Gen}(1^n)$, a random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_k(m_b)$ is given to A .
3. A outputs b'
4. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Computational Indistinguishability against eavesdropper

Definition (EAV security): a private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is **EAV-secure**, if for all **probabilistic polynomial time** adversaries A , there is a **negligible function** negl such that for all n ,

$$\Pr[\text{PrivK}_{A,\Pi}^{eav}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

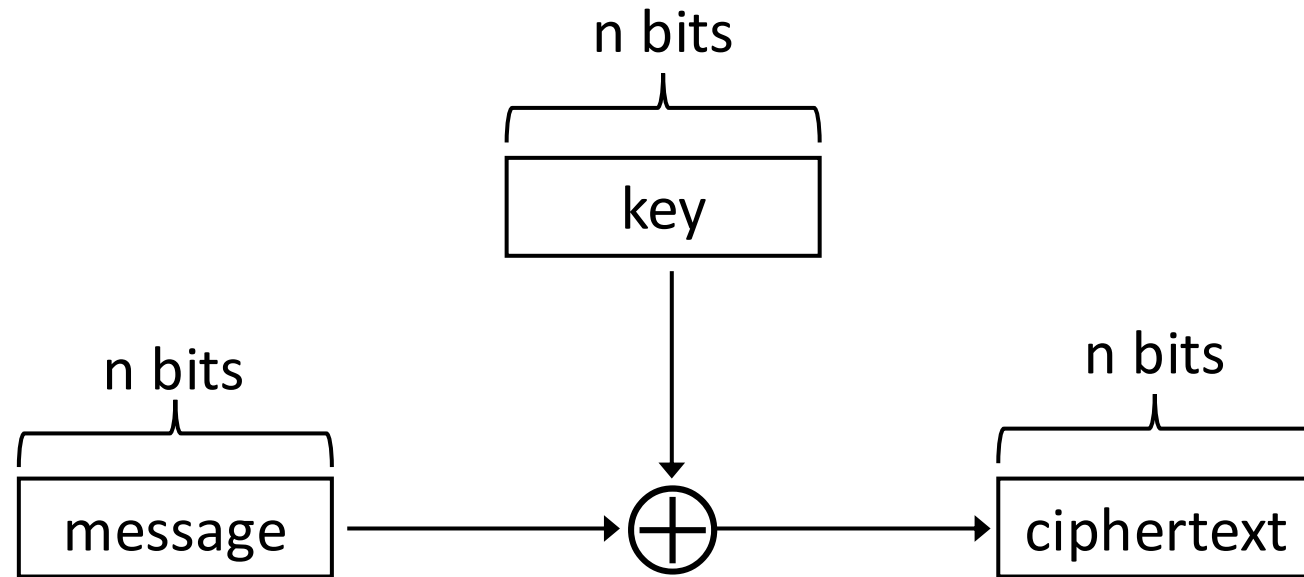
More on the definition

- Honest encryption and decryption $O(n^c)$. E.g., 60^2 CPU cycles for $n=60$
- Adversary may break it in time $O(2^n) = 2^{60}$ CPU cycles
- PPT adversary may break it with probability $O(2^{-n}) = \frac{\text{poly}(n)}{2^{60}}$
- If not enough, increase the security parameter n from 60 to 100

Next

1. Construction to achieve computational EAV security where key is shorter than the message
2. Security for multiple encryptions

One-time pad



Pseudorandom Generator (PRG)

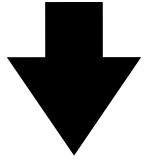
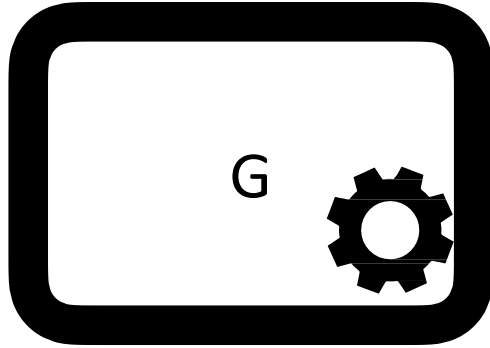
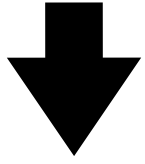
A PRG is a function $G : \{0,1\}^n \rightarrow \{0,1\}^{n+s}$

Can we turn a short random string into a long random string?

No, this is provably impossible

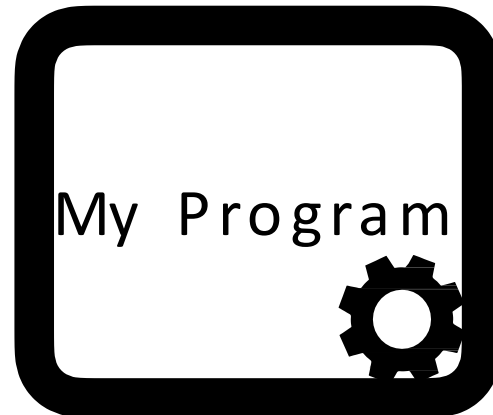
PRG: turning a short random string into a long string that **looks** random

01101010



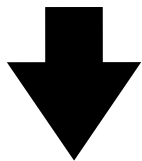
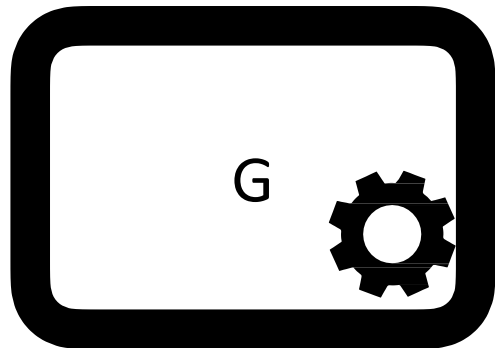
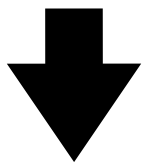
101101111011001

111011000110110

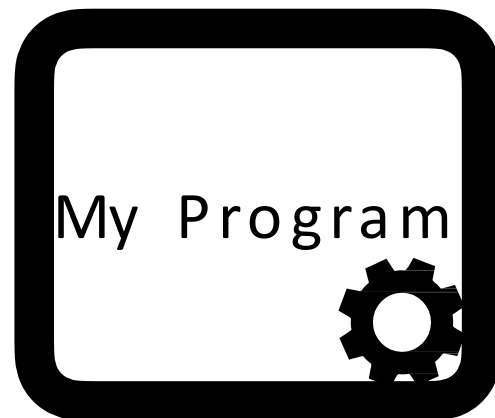
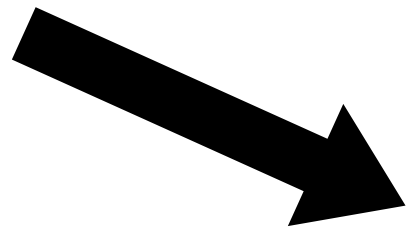


Slide credit to David Heath

01101010



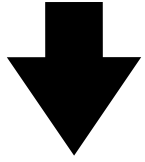
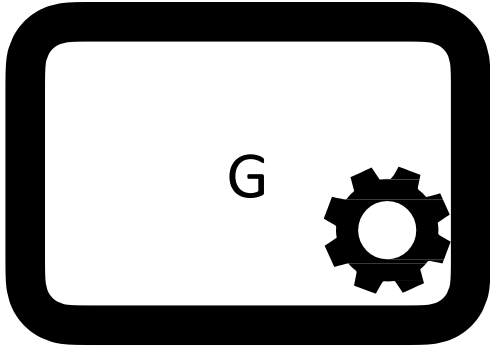
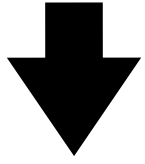
101101111011001



111011000110110

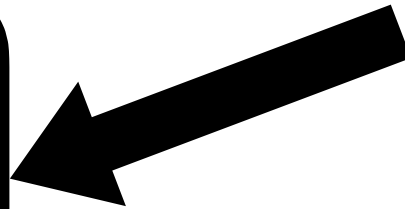
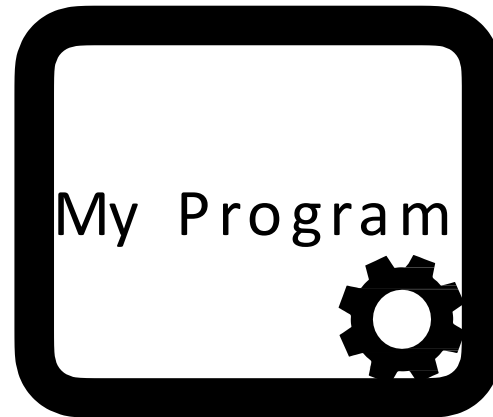


01101010

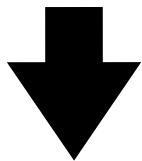
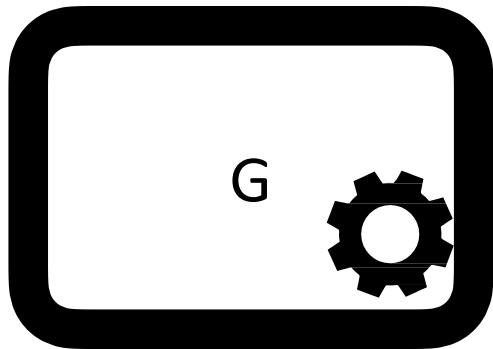
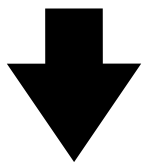


101101111011001

111011000110110



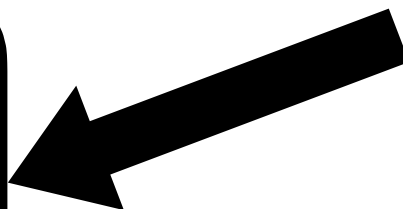
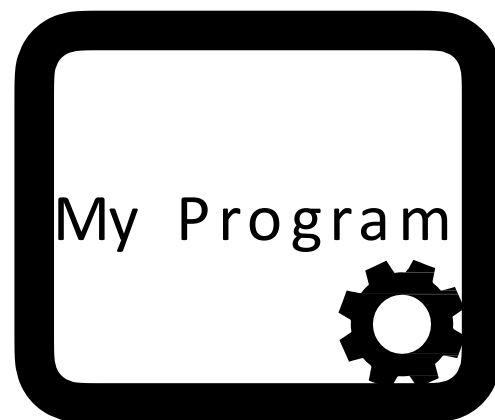
01101010



101101111011001

**G is a PRG if *no* program can
reliably win this game**

111011000110110



REAL/FAKE

Definition

A PRG is a function $G : \{0,1\}^n \rightarrow \{0,1\}^{n+s}$

Pseudorandomness: for any PPT algorithm D , there is a negligible function negl such that

$$|\Pr[D(G(k)) = 1] - \Pr[D(r) = 1]| \leq \text{negl}(n)$$

For uniformly random $k \in \{0,1\}^n$ and $r \in \{0,1\}^{n+s}$

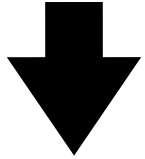
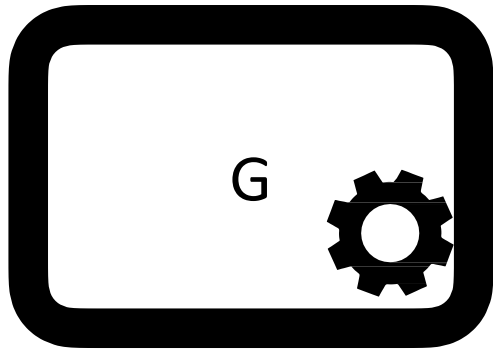
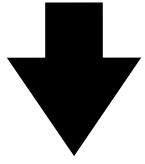
Example that is not a PRG

- $G : \{0,1\}^n \rightarrow \{0,1\}^{n+1}$ by computing the XOR of the first n bits
- D : if $r_{n+1} = \bigoplus_{i=1}^n r_i$, output 1, otherwise, output 0.

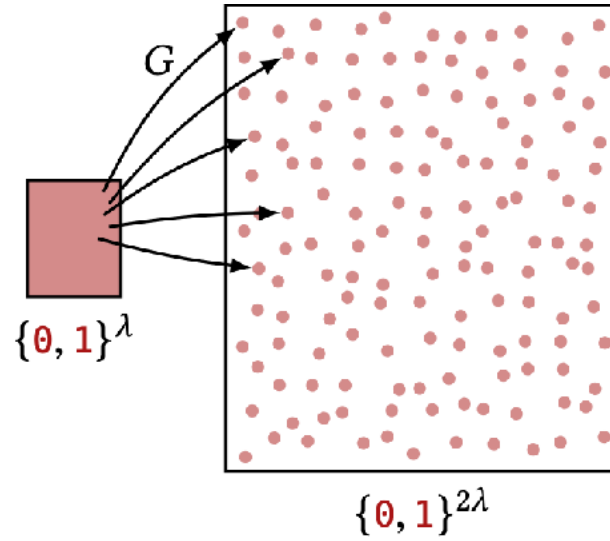
Brute-force search

- Distinguisher:
 1. precompute $G(s)$ for all possible $s \in \{0,1\}^n$
 2. If the string is in the table, output 1, otherwise output 0.
- What's $\Pr[D(G(k)) = 1]$?
- What's $\Pr[D(r) = 1]$ if $G : \{0,1\}^n \rightarrow \{0,1\}^{n+1}$?
- What's $\Pr[D(r) = 1]$ if $G : \{0,1\}^n \rightarrow \{0,1\}^{2n}$?

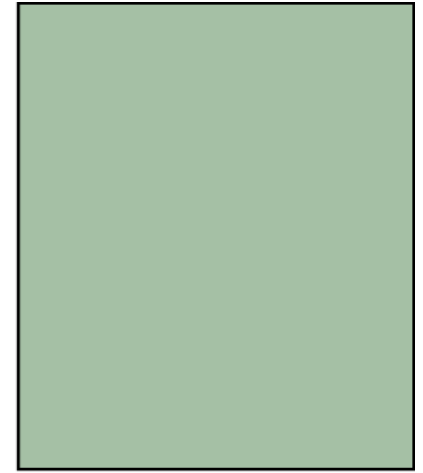
01101010



101101111011001



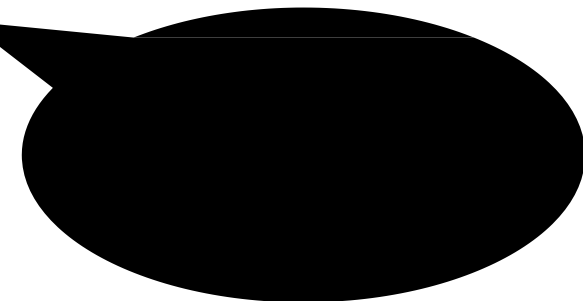
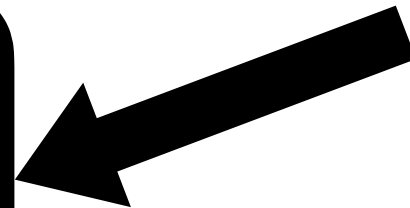
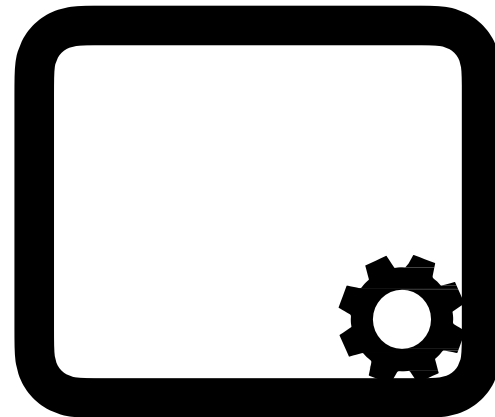
pseudorandom distribution



$\{0, 1\}^{2\lambda}$

uniform distribution

111011000110110



Existence of PRG

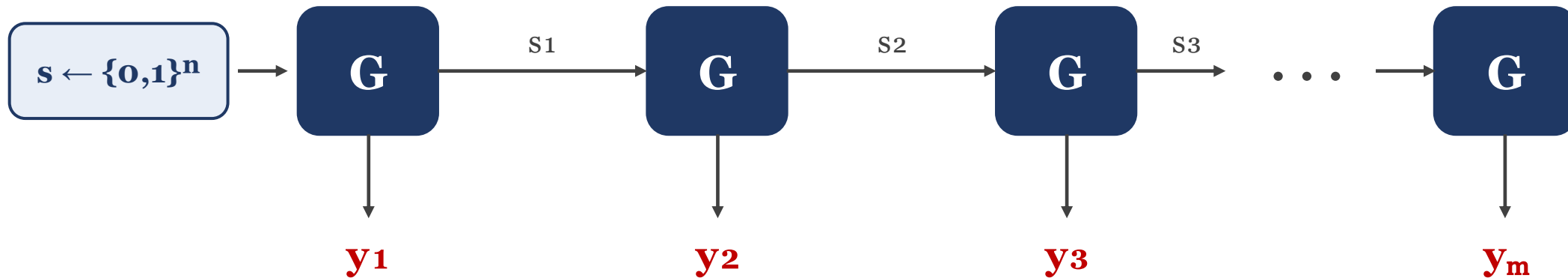
- We believe that PRGs exist – assumption
- Construction from NP hard problems (3-SAT)
 1. One-bit PRG to m-bit PRG
 2. One-way function $\Leftrightarrow$ PRG

Foundations of Cryptography, Volume 1: Basic Tools (2001)

If PRG exists, then $P \neq NP$

From 1-bit PRG to m-bit PRG

One-bit PRG $G : \{0,1\}^n \rightarrow \{0,1\}^{n+1} \Rightarrow$ m-bit PRG $G' : \{0,1\}^n \rightarrow \{0,1\}^m$



$$G'(s_0) = y_1 y_2 y_3 \dots y_m$$

Each stage: one PRG call keeps the n -bit state and emits one output bit. m calls $\Rightarrow m$ output bits, for any polynomial m .

EAV-security from PRG

Construction:

- $\text{Gen}(1^n)$: choose uniform $k \in \{0,1\}^n$ and output it as the key
- $\text{Enc}_k(m)$: on input a message $m \in \{0,1\}^{n+s}$, compute

$$c = G(k) \oplus m$$

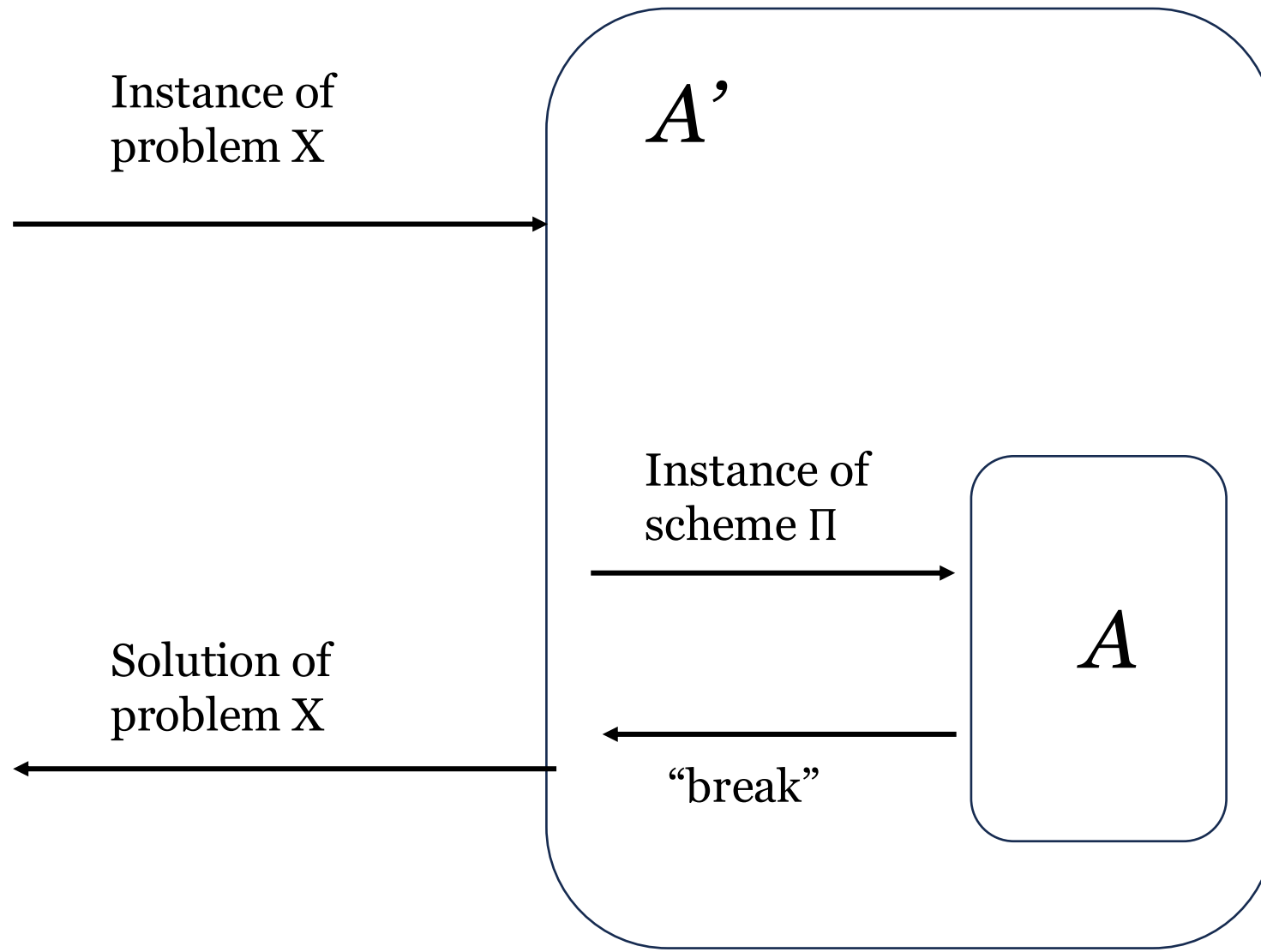
- $\text{Dec}_k(c)$: on input a ciphertext $c \in \{0,1\}^{n+s}$, compute

$$m = G(k) \oplus c$$

EAV-security from PRG

- Theorem: if G is a PRG, then the construction is EAV secure.

Proof by reduction



- X is the hard problem in the assumption
- Polynomial time reduction
- Non-negligible success probability

Indistinguishability experiment

Experiment $\text{PrivK}_{A,\Pi}^{\text{eav}}(n)$:

1. Adversary A is given input 1^n , and outputs two messages $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1|$
2. A key k is generated using $\text{Gen}(1^n)$, a random bit $b \in \{0,1\}$ is chosen. $c = \text{Enc}_k(m_b)$ is given to A .
3. A outputs b'
4. Experiment outputs 1 if $b=b'$, and 0 otherwise.

Definition (EAV security): a private-key encryption $\Pi = (\text{Gen}, \text{Enc}, \text{Dec})$ is **EAV-secure**, if for all **probabilistic polynomial time** adversaries A , there is a **negligible function** negl such that for all n ,

$$\Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] \leq \frac{1}{2} + \text{negl}(n)$$

- Suppose there is A breaks the EAV security

$$\Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] > \frac{1}{2} + \text{negl}(n)$$

- Distinguisher D with input 1^n and a string w of $(n+s)$ bits
 1. Run A to get $m_0, m_1 \in \mathcal{M}$ s.t. $|m_0| = |m_1| = n + s$
 2. Choose a random bit $b \in \{0,1\}$, set $c = w \oplus m_b$
 3. Give c to A and obtain output b' . Output 1 if $b'=b$, 0 otherwise

If w is a random string,

$$\Pr[D(w) = 1] = \Pr[\text{PrivK}_{A,\text{OTP}}^{\text{eav}}(n + s) = 1] = \frac{1}{2}$$

If $w = G(k)$,

$$\Pr[D(w) = 1] = \Pr[\text{PrivK}_{A,\Pi}^{\text{eav}}(n) = 1] > \frac{1}{2} + \text{negl}(n)$$